TASK 18 — MATLAB EXTENDED RESEARCH REPORT

GEMSTAR Camera Perception | Python/OpenCV Reference + MATLAB Image Processing Toolbox + Integer/FPG■-Oriented Experiments

This extended report retains the original Task 18 research evidence, and adds the completed MATLAB experiments 18.11–18.16. It does not replace or alter the original PDF.

Executive summary

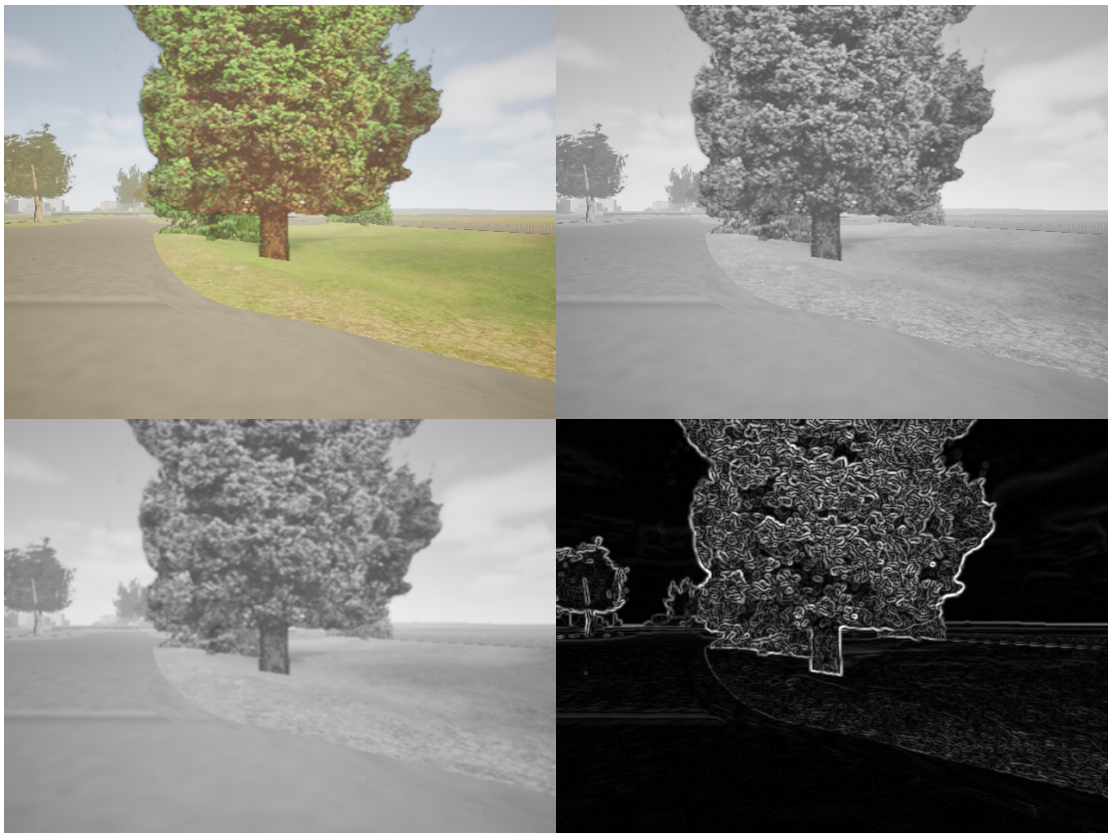
Twelve 640 × 480 GEMSTAR Camera frames were extracted from a 466-message SensorView recording. The previous work established OpenCV preprocessing, MATLAB numerical validation, provisional YOLO11n/YOLO11s detection evaluation, and a 120-measurement repeated PC inference benchmark. The extension implements three MATLAB preprocessing stages on all selected frames, evaluates image statistics, measures MATLAB CPU operator time, and develops integer grayscale and Sobel reference algorithms. All reported quality metrics describe this selected dataset and are not an independently validated sensor-reliability score.

1. Original recorded-camera dataset and visual examples

Source recording: 20261027T224313Z_sv_380_X_466_GEMSTAR.osi. Selected frame IDs: 0, 50, 100, 150, 200, 250, 300, 350, 400, 423, 424, 450. Camera payload interpreted as RGB, 640 × 480 × 3 (921,600 bytes), based on visual inspection. The original run has two previously identified frame-timing regimes.



Original experiment: all twelve representative Camera frames.



Original Python/OpenCV Camera processing pipeline.

2. Original YOLO detection and reference evaluation

Two pretrained YOLO11 models were tested on PC CPU with imgsz=640, confidence=0.25, NMS IoU=0.7 and class-agnostic NMS. Reference annotations span just five provisional target objects across 12 selected frames. Detection confidence is not a calibrated reliability probability. Errors such as cars classified as unrelated categories motivate future cross-sensor checking.

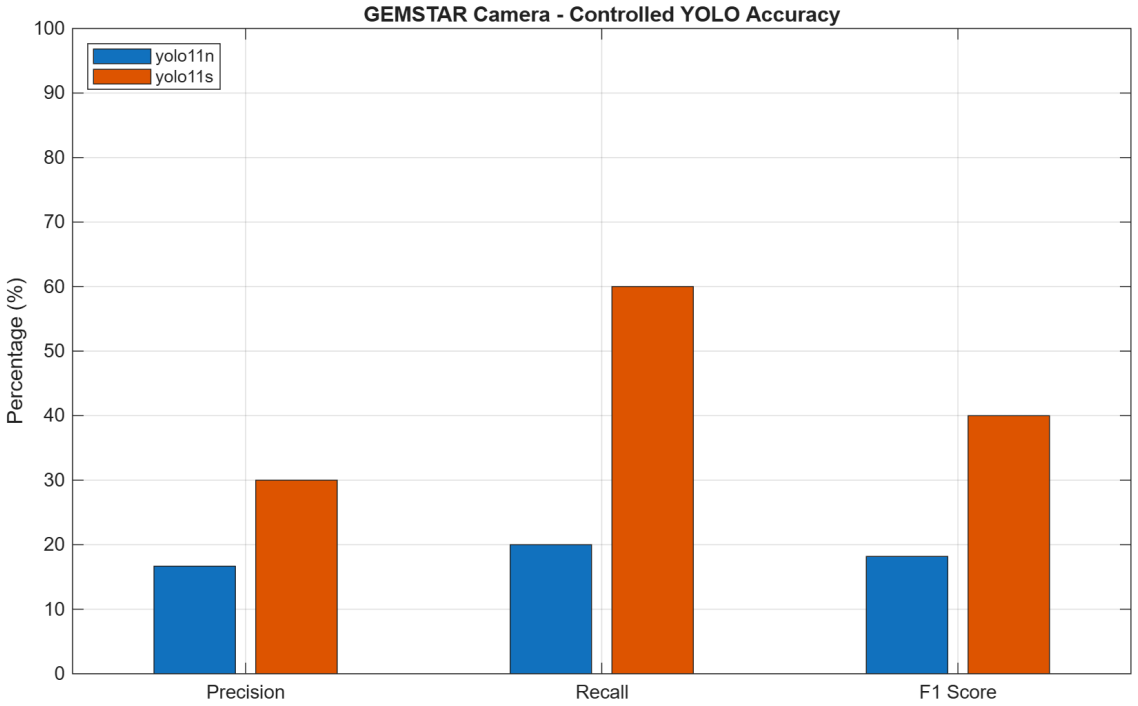


Initial detection examples including visually observed classification errors.



Provisional manually annotated frames; annotations have not undergone independent review.

Model	TP	FP	FN	Precision	Recall	F1
YOLO11 Nano	1	5	4	16.67%	20.00%	18.18%
YOLO11 Small	3	7	2	30.00%	60.00%	40.00%

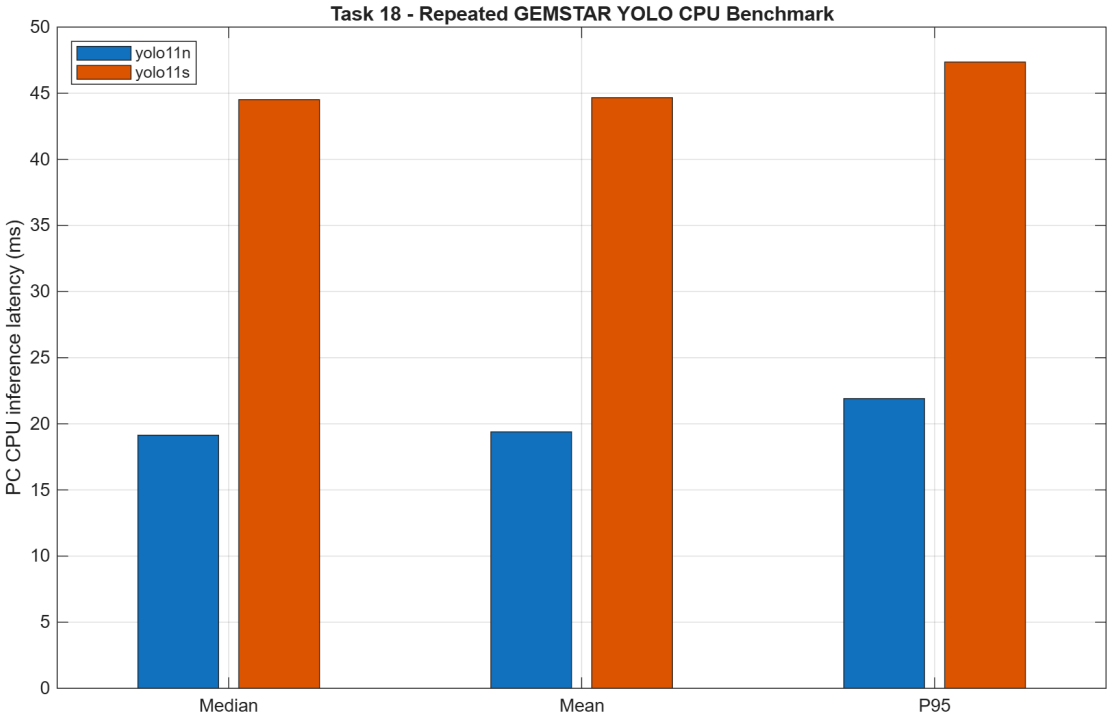


Python/MATLAB independently verified controlled accuracy comparison.

3. Original repeated desktop CPU inference benchmark

Model	Measurements	Median	Mean	P95	Std. dev.
YOLO11 Nano	60	19.137 ms	19.392 ms	21.911 ms	1.497 ms
YOLO11 Small	60	44.512 ms	44.664 ms	47.356 ms	1.887 ms

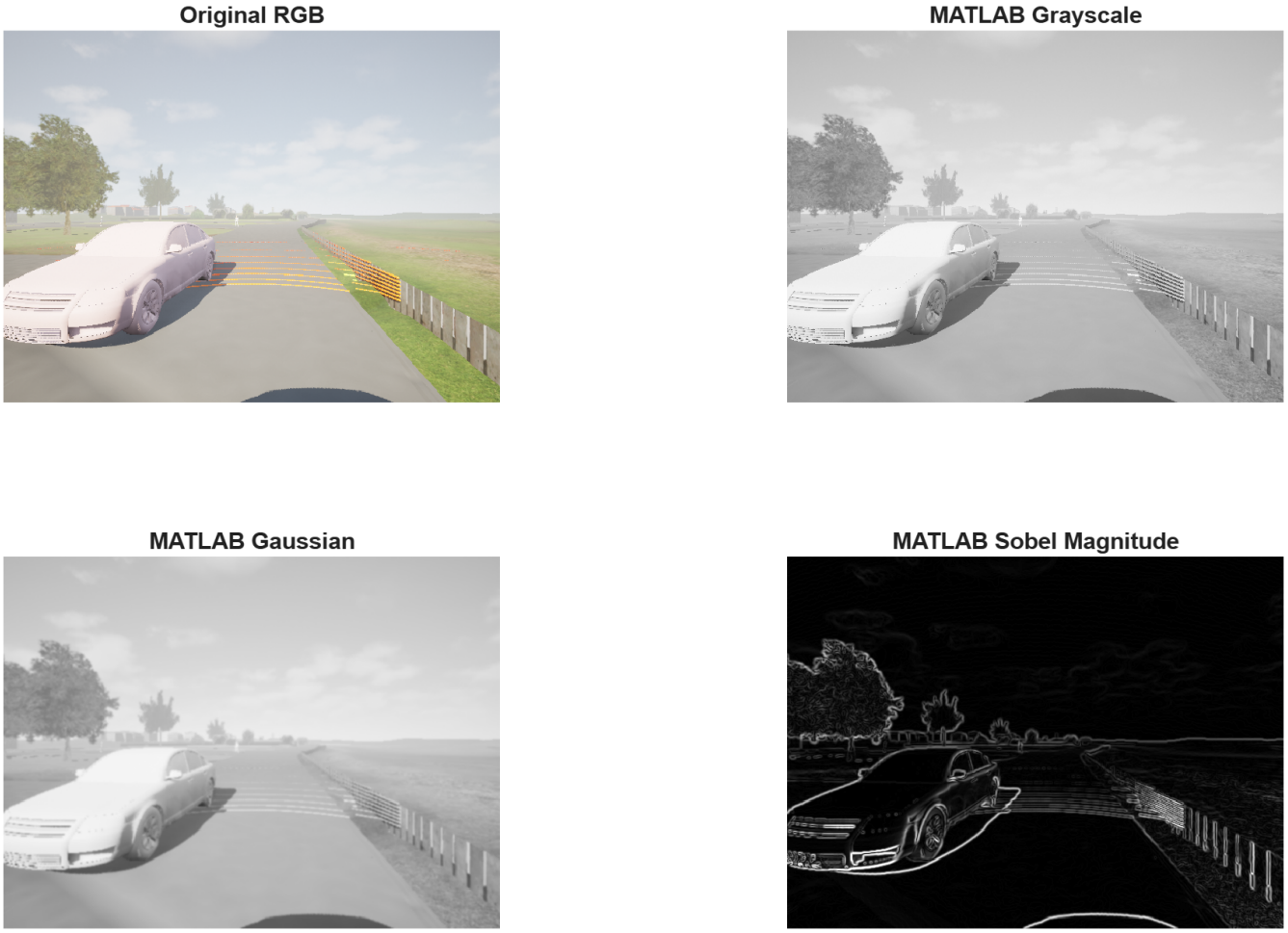
The Small model median was approximately 2.33 times Nano. These values refer to PC CPU model inference, not total pipeline duration, FPGA execution, or board-level energy.



Repeated CPU inference comparison, independently recalculated in MATLAB.

4. Task 18.11 — MATLAB toolbox-based Camera preprocessing

MATLAB R2026b Image Processing Toolbox was installed and functionally checked using fspecial, imfilter and imshowpair. The production MATLAB script processed the twelve saved GEMSTAR RGB images with rgb2gray, a 5×5 Gaussian kernel (sigma 1) via imfilter with replicate boundaries, and imgradient Sobel magnitude. Thirty-six processed images were saved, three per frame.



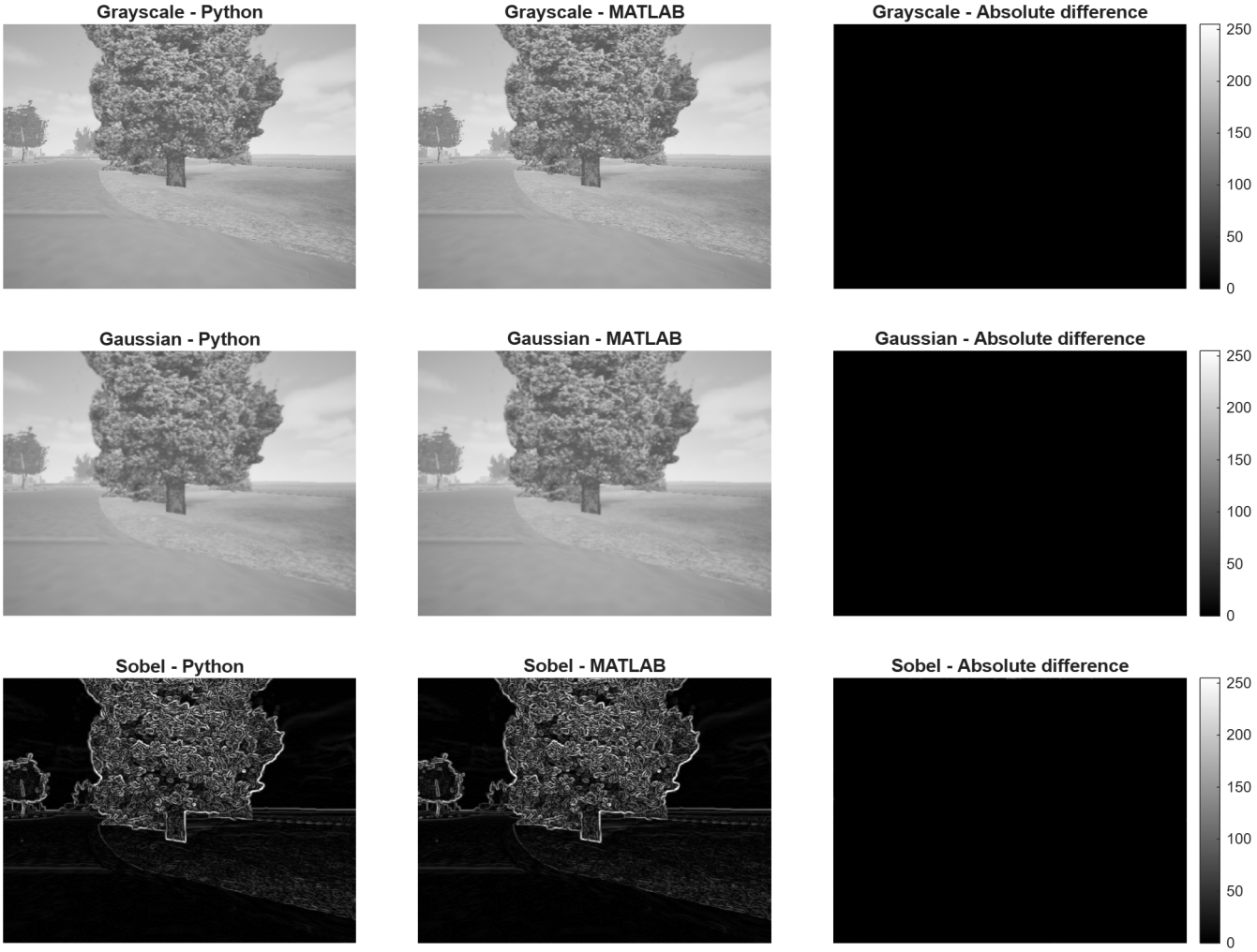
Frame 150, MATLAB toolbox pipeline: RGB, grayscale, Gaussian and Sobel.

Frame	MeanGray	MeanBlur	MeanEdge
0	164.8253	164.8260	24.8573
50	189.0509	189.0514	12.0390
100	190.7995	190.8002	10.9461
150	192.8147	192.8142	13.3660
200	191.4203	191.4203	9.5168
250	190.4923	190.4927	10.4334
300	193.7392	193.7414	12.0897
350	174.4095	174.4092	14.4484
400	188.7938	188.7936	14.2819
423	193.3888	193.3901	12.8896
424	193.3785	193.3791	12.8299
450	170.0849	170.0865	13.5531

5. Task 18.12 — MATLAB versus Python/OpenCV (frame 0)

Comparison is performed at the saved uint8 pixel level; small differences can arise from filter kernels, arithmetic rounding, Sobel scaling and border handling. Visual similarity does not imply bit-exact equality. The large maximum Sobel deviation merits attention even though average error is small.

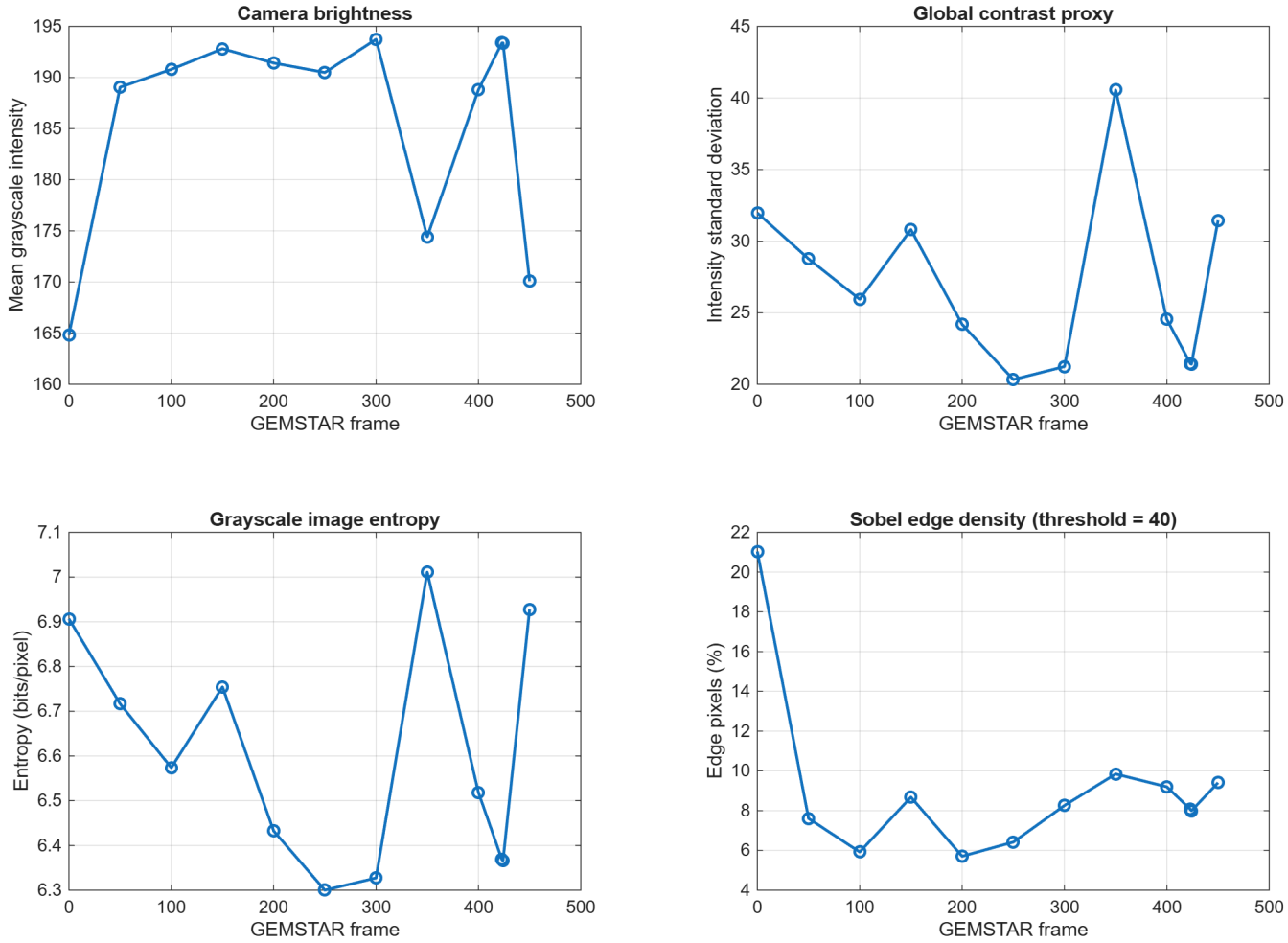
Operation	MAE	RMSE	Maximum error	Exact-match %
Grayscale	0.0006	0.0241	1.0000	99.9417
Gaussian	0.0178	0.2101	16.0000	98.5990
Sobel	0.1402	1.1303	100.0000	92.7591



Three-stage saved-output comparison, showing MATLAB, Python and absolute pixel differences.

6. Task 18.13 — Image-content metrics on 12 frames

Measurements: grayscale brightness mean, global intensity standard deviation (contrast proxy), Shannon entropy, average Gaussian-induced pixel change and Sobel edge-pixel percentage at a fixed threshold of 40. These indicate scene-content variations. They do NOT independently identify fog severity or Camera sensor failure.



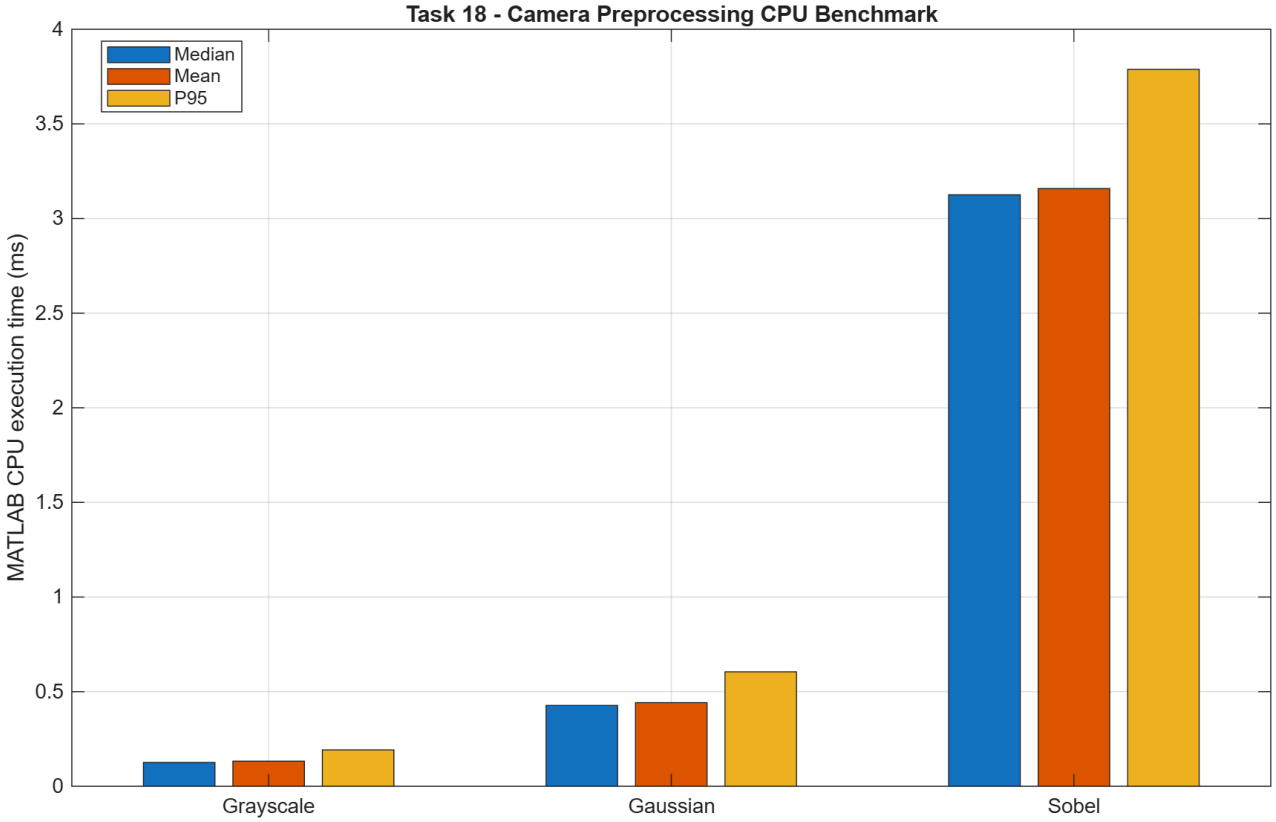
Brightness, contrast proxy, grayscale entropy and Sobel edge density by selected frame.

Frame	Brightness	Contrast	ImageEntropy	GaussianMAE	EdgeDensity
0	164.8253	31.9715	6.9066	2.4452	21.0345
50	189.0509	28.7627	6.7166	1.6329	7.6090
100	190.7995	25.9363	6.5738	1.5086	5.9199
150	192.8147	30.8108	6.7545	1.8255	8.6794
200	191.4203	24.1828	6.4324	1.3013	5.7051
250	190.4923	20.3294	6.3001	1.4770	6.4189
300	193.7392	21.2359	6.3273	1.7419	8.2598
350	174.4095	40.5656	7.0112	1.8677	9.8431
400	188.7938	24.5356	6.5184	1.9623	9.1982
423	193.3888	21.4455	6.3682	1.9044	8.0677
424	193.3785	21.3982	6.3659	1.8733	7.9932
450	170.0849	31.4485	6.9278	1.5916	9.4059

7. Task 18.14 — MATLAB preprocessing timing benchmark

Each operation was measured ten times on each of twelve images after three warm-up executions. The 360 recorded MATLAB PC timing values exclude file reading and writing. They are not directly comparable to neural-network inference or ARM-FPGA timings as equivalent workloads.

Operation	Measurements	MedianMs	MeanMs	P95Ms	StdDevMs
Grayscale	120	0.1259	0.1329	0.1924	0.0429
Gaussian	120	0.4269	0.4419	0.6050	0.0910
Sobel	120	3.1254	3.1585	3.7887	0.3259

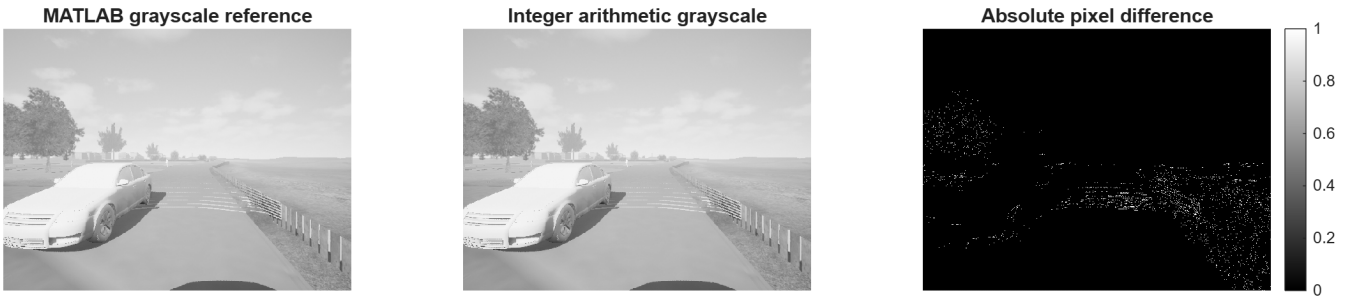


Measured desktop MATLAB CPU processing latencies; Sobel is the most expensive of the three operators in this implementation.

8. Task 18.15A — Integer grayscale for streaming hardware

An integer weighted approximation uses Y = (77R + 150G + 29B + 128) >> 8. Input samples remain 8-bit, but accumulation uses uint32 to prevent overflow. This is a coefficient approximation to RGB grayscale; it is not identical to MATLAB rgb2gray. Distinct bit widths should be tested when converting this algorithm into HDL/HLS.

Frames	Mean MAE	Worst MAE	Worst max error	Mean exact match
12	0.0114	0.0223	1.0000	98.8620%



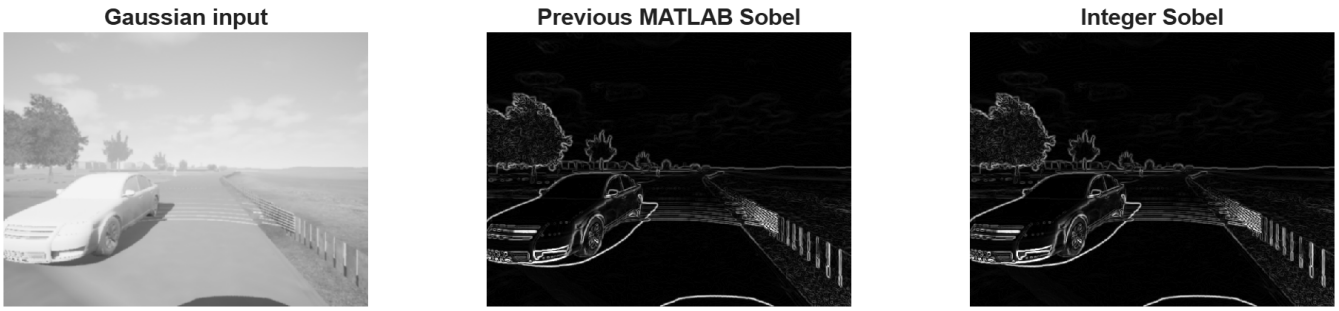
Reference versus coefficient-quantized grayscale and absolute difference, frame 150.

9. Task 18.15B — Integer signed Sobel gradients

Signed int32 arithmetic was used for Sobel Gx and Gy; this agrees exactly with a floating-point reference using the same coefficients, image, boundaries, and final hypot/round/clamp operation. Thus the gradients are integer but the final L2 magnitude is not yet integer-only. Standard 3×3 Sobel worst-case single-axis gradient is ±1020, so 11 signed bits suffice for

each gradient before magnitude combination.

Frames	Mean MAE	Min exact match	Max |Gx|	Max |Gy|
12	0.0000	100.0000%	337.0000	359.0000



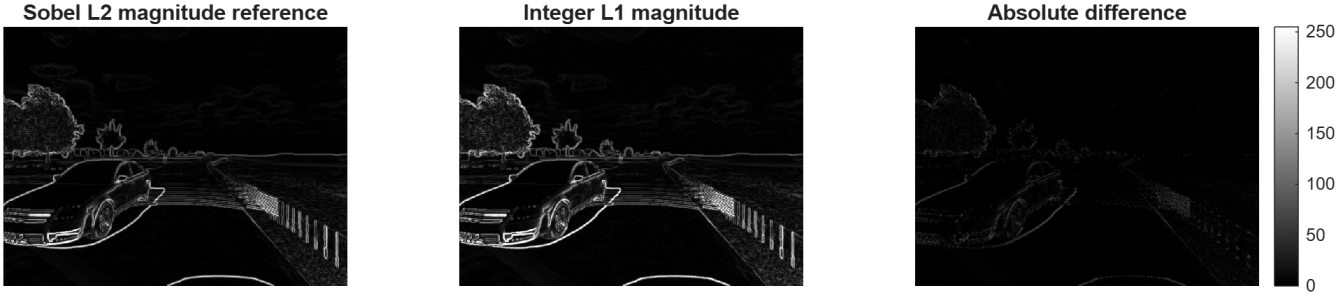
Frame 150: smoothed Camera input, saved Sobel and signed-integer gradient implementation.

10. Task 18.16 — Fully integer L1 Sobel magnitude approximation

The new magnitude approximation is min(255, |Gx|+|Gy|), replacing sqrt(Gx²+Gy²). The approximation changes output values, can amplify diagonal edges, and may saturate at 255 more often than L2. Pixel-level differences are measured directly against the L2 reference using the SAME integer Sobel gradients. Image similarity does not by itself demonstrate lower FPGA resource usage or energy savings.

Frames	Mean MAE	Worst max error	Mean exact %	Mean sat. %
12	3.1995	74.0000	32.8881	0.1017

Frame	MAE	Max error	Exact %	Saturation %
0	6.3342	74.0000	24.2715	0.2357
50	2.7099	74.0000	34.7536	0.0931
100	2.3656	73.0000	36.9993	0.0794
150	2.9540	74.0000	39.3835	0.3154
200	2.0734	73.0000	41.6273	0.0322
250	2.4486	74.0000	34.0075	0.0010
300	2.8554	71.0000	34.7669	0.0055
350	3.5618	74.0000	28.9867	0.0576
400	3.4558	74.0000	26.6019	0.1029
423	3.1012	73.0000	27.8470	0.0195
424	3.0926	74.0000	27.5514	0.0195
450	3.4413	74.0000	37.8607	0.2581



Frame 150: exact L2 and integer L1 Sobel magnitude with absolute error image.

11. Reproducibility and important limitations

• All model-accuracy claims are exploratory: only 12 selected frames and five provisional manual target-object annotations, with no independent reference-label review.

• MATLAB/OpenCV comparisons may depend on border padding, rounding, saved 8-bit saturation and gradient scaling; identical numerical output must be explicitly engineered if required.

• The MATLAB preprocessing latency benchmark uses tic/toc on a desktop and is affected by cache, JIT, processor load and implementation; it is not a comparison to OpenCV until both use matched test methodology.

• Integer grayscale and Sobel calculations are numerical reference experiments, not resource, power, timing or energy measurements on a synthesized FPGA.

• RGB input into pretrained YOLO remains distinct from the grayscale/Sobel branch; no detection improvement from preprocessing has been demonstrated.

• No claim of validated adaptive CPU-FPGA scheduling, energy savings or end-to-end real-time performance is supported by Task 18.

12. Task 18 research archive

01_Graphs: 15 local files.

02_CSV_Data: 27 local files.

03_Configurations: 2 local files.

04_Code: 29 local files.

05_Raw_Data_References: 0 local files.

06_Research_Findings: 1 local files.

07_Research_Report: 1 local files.

08_Extracted_Images: 116 local files.

09_GroundTruth_Annotations: 13 local files.

Next work: preserve a deterministic hardware input-output test-vector format and, in subsequent tasks, profile corresponding operators on actual ARM/FPGA hardware with measured end-to-end timing, transfer overhead, resource utilization and energy.